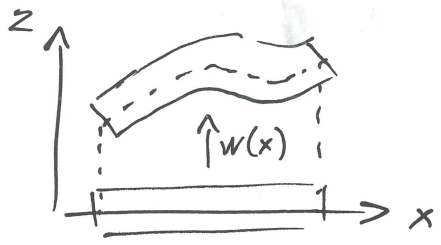
Övning 9 - Elastiska linjens ekvation

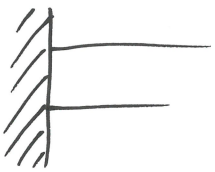


$w(x)$ - Medellinjens förskjutning

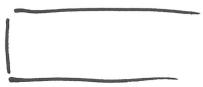
$$EIw''''(x) = q(x) \quad [FS.\ 6.36]$$

$w(x)$ - Förskjutning	$z(x)$
$w'(x)$ - Vinkeländring	$\Theta(x)$
$-EIw''(x)$ - Momentet	$M(x)$
$-EIw'''(x)$ - Tvärkraften	$T(x)$

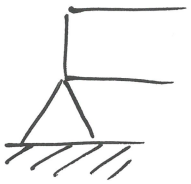
Randvillkor



$w = 0$
$\Theta = 0$
$T \neq 0$
$M \neq 0$



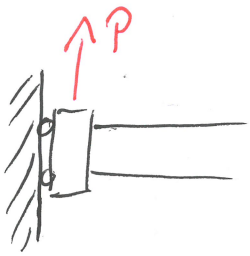
$w \neq 0$
$\Theta \neq 0$
$T = 0$
$M = 0$



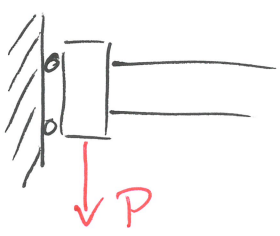
$w = 0$
$\Theta \neq 0$
$T \neq 0$
$M = 0$



$w = 0$
$\Theta \neq 0$
$T \neq 0$
$M = M_0$



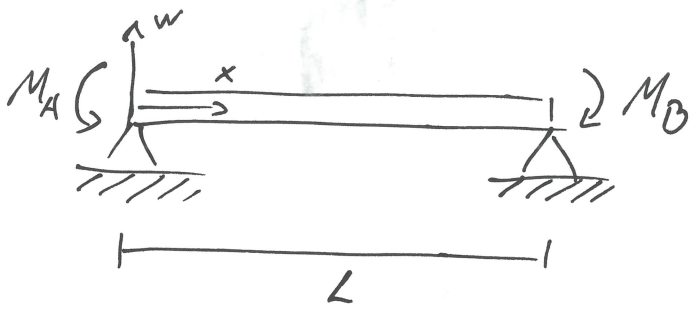
$w \neq 0$
$\Theta = 0$
$T = -P$
$M \neq 0$



$w \neq 0$
$\Theta = 0$
$T = P$
$M \neq 0$

2.4.97

Balklängd L
E-modul E



Sökt: Utböjningen $w(x)$

① Ställ upp elastiska linjen

$$EI\,w''''(x) = q(x) \qquad q(x) = 0$$

$$w''''(x) = 0$$

$$w'''(x) = C_1$$

$$w''(x) = C_1 x + C_2$$

$$w'(x) = C_1 \frac{x^2}{2} + C_2 x + C_3$$

$$w(x) = C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4$$

② Rand-villkor

- $w(0) = \boxed{C_4 = 0}$
- $M(0) = M_A \qquad M(x) = -EI\,w''(x)$

$$M_A = -EI\,(C_1 \cdot 0 + C_2) \Rightarrow \boxed{C_2 = -\frac{M_A}{EI}}$$

- $M(L) = M_B$

$$M_B = -EI\left(C_1 \cdot L - \frac{M_A}{EI}\right) \Rightarrow \boxed{C_1 = \frac{M_A - M_B}{LEI}}$$

- $w(L) = 0 = C_1 \frac{L^3}{6} + C_2 \frac{L^2}{2} + C_3 L$

$$0 = \left(\frac{M_A - M_B}{LEI}\right)\frac{L^3}{6} - \frac{M_A}{EI}\frac{L^2}{2} + C_3 \cdot L$$

$$\Rightarrow \boxed{C_3 = \frac{(2M_A + M_B)\,L}{6EI}}$$

2.4:97 del 2

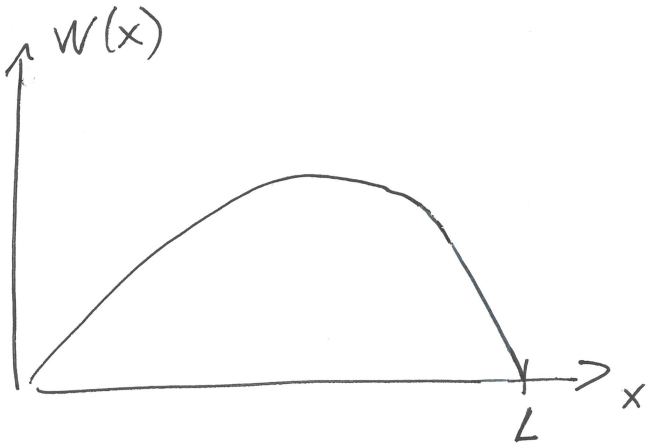
$$w(x) = C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4$$

$$C_1 = \frac{M_A - M_B}{6EI} \qquad C_2 = -\frac{M_A}{EI} \qquad C_3 = \frac{(2M_A + M_B)L}{6EI} \qquad C_4 = 0$$

$$w(x) = \frac{M_A - M_B}{6LEI} x^3 - \frac{M_A}{2EI} x^2 + \frac{(2M_A + M_B)L}{6EI} x$$

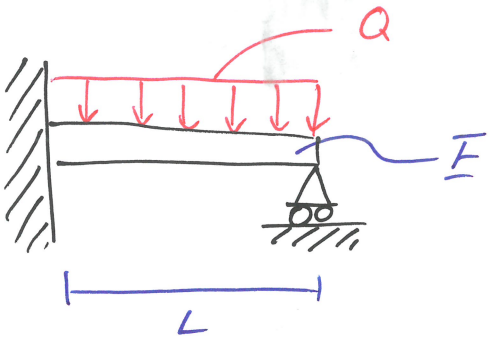
Exempel: $M_A = 1$ $M_B = 2$

$$w(x) = -\frac{1}{6LEI} x^3 - \frac{1}{2EI} x^2 + \frac{2L}{3EI} x$$

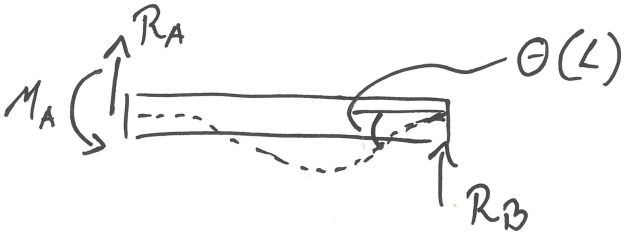


2.4.117

Sökt: Reaktionskrafter R_A R_B
Reaktionsmoment M_A
Vinkel ändring vid stöd $\theta(L)$



① Frilägging



Randvillkor

(A) $\begin{cases} w(0) = 0 & \text{I} \\ \theta_A = w'(0) = 0 & \text{II} \end{cases}$

(B) $\begin{cases} w(L) = 0 & \text{III} \\ M(L) = -EIw''(L) = 0 & \text{IV} \end{cases}$

② Elastiska linjen

$$EIw''''(x) = q(x) \qquad q(x) = -\frac{Q}{L}$$

$$EIw'''(x) = -\frac{Q}{L}$$

$$EIw'''(x) = -\frac{Q}{L}x + C_1$$

$$EIw''(x) = -\frac{Q}{L}\frac{x^2}{2} + C_1x + C_2$$

$$EIw'(x) = -\frac{Q}{L}\frac{x^3}{6} + C_1\frac{x^2}{2} + C_2x + C_3$$

$$\boxed{EIw(x) = -\frac{Q}{L}\frac{x^4}{24} + C_1\frac{x^3}{6} + C_2\frac{x^2}{2} + C_3x + C_4}$$

I $\Rightarrow C_4 = 0$ $\qquad$ II $\Rightarrow C_3 = 0$

III $\Rightarrow QL^3 = 4C_1L^3 + 12C_2L =$ $\qquad$ IV $\Rightarrow -\frac{QL}{2} + C_1L + C_2 = 0$

$$C_1 = \frac{5}{8}Q \qquad C_2 = -\frac{QL}{8}$$

2.4.117 del 2

④ Tvärkrafter

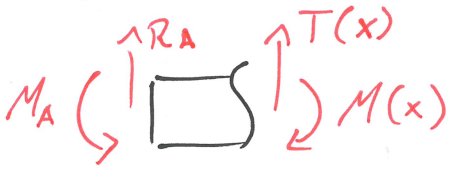
$$T(x) = -EIw'''(x)$$
$$M(x) = -EIw''(x)$$

Vinkeländring

$$\theta = w'(x)$$

$$\boxed{\theta(L)} = w'(L) = \frac{1}{EI}\left(-\frac{QL^2}{6} + \frac{5QL^2}{16} - \frac{QL^2}{8}\right) = \boxed{\frac{QL^2}{48EI}}$$

Snitta

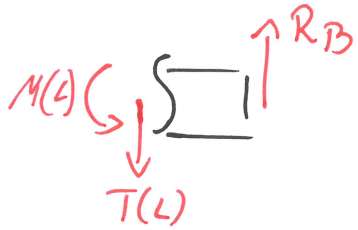


$$R_A = -T(0) = EIw'''(0)$$

$$\boxed{R_A = \frac{5}{8}Q}$$

$$M_A = M(0) = -EIw''(0)$$

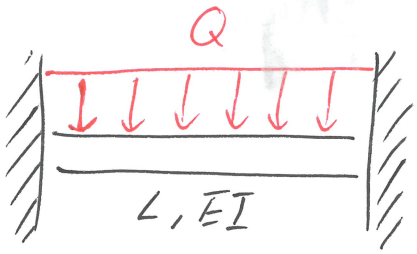
$$\boxed{M_A = \frac{QL}{8}}$$



$$R_B = T(L) = -EIw'''(L)$$

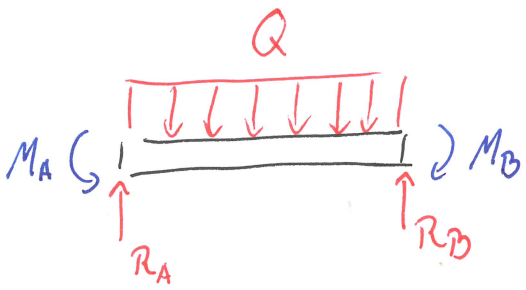
$$\boxed{R_B = \frac{3}{8}Q}$$

2.4:120



Sökt: Ändmomenten M_A, M_B
Nedböjningen i mitten av stången $w(\frac{L}{2})$

① Frilägg ning och bestäm randvillkor



Fast inspänd i båda ändar

$w(0) = w(L) = 0$ (Förskjutning)

$w'(0) = w'(L) = 0$ (Vinklar)

② Elastiska linjen

$$EIw''''(x) = q \qquad q = -\frac{Q}{L}$$

Integrera fyra steg

$$EIw(x) = -\frac{Q}{24L}x^4 + \frac{C_1}{6}x^3 + \frac{C_2}{2}x^2 + C_3x + C_4$$

Randvillkor

$$w(0) = 0 \Rightarrow C_4 = 0$$

$$w'(0) = 0 \Rightarrow C_3 = 0$$

$$w(L) = 0 \Rightarrow -\frac{Q}{24L}L^4 + \frac{C_1}{6}L^3 + \frac{C_2}{2}L^2 = 0$$

$$w'(L) = 0 \Rightarrow -\frac{Q}{6L}L^3 + \frac{C_1}{2}L^2 + C_2L = 0$$

Två obekanta två ekvationer

$$\Rightarrow \quad C_1 = \frac{Q}{2} \qquad C_2 = -\frac{QL}{12}$$

2.4:120 del 2

$$EIw(x) = -\frac{Q}{24L}x^4 + \frac{Q}{12}x^3 - \frac{QL}{24}x^2$$

Från formelsamling $M = -EIw''$ [FS. 6.39]

$$M(0) = -EIw''(0) = \frac{QL}{12}$$

$$M(L) = -EIw''(L) = \frac{Q}{2L}L^2 - \frac{Q}{2}L + \frac{QL}{12} = \frac{QL}{12}$$

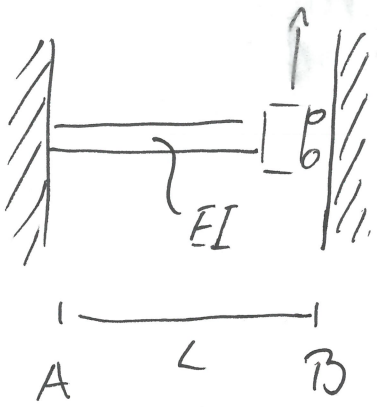
$$\boxed{M_A = \frac{QL}{12}} \quad \boxed{M_B = \frac{QL}{12}}$$

Nedböjning i mitten

$$w\left(\frac{L}{2}\right) = \frac{1}{EI}\left(-\frac{Q}{24L}\left(\frac{L}{2}\right)^4 + \frac{Q}{12}\left(\frac{L}{2}\right)^3 - \frac{QL}{24}\left(\frac{L}{2}\right)^2\right)$$

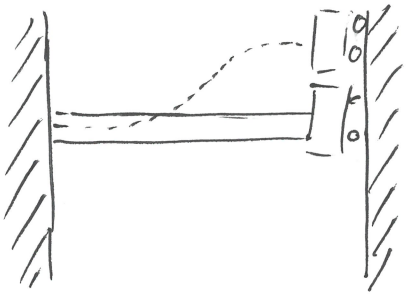
$$\boxed{w\left(\frac{L}{2}\right) = -\frac{QL^3}{384EI}}$$

2.4.123



Sökt: Förskjutningen vid punkt B $w(L)$

① Bestäm randvillkor



$w(0) = 0$ (1)

$w'(0) = 0$ (2)

$T(L) = -EIw'''(L) = P$ (3)

$w'(L) = 0$ (4)

② Elastiska linjen

$$w(x) = \frac{C_1 x^3}{6} + \frac{C_2 x^2}{2} + C_3 x + C_4$$

$w(0) = 0 \Rightarrow C_4 = 0$ Randvillkor (1)

$w'(0) = 0 \Rightarrow C_3 = 0$ Randvillkor (2)

$-EIw'''(L) = -EIC_1 = P \Rightarrow C_1 = -\frac{P}{EI}$ Randvillkor (3)

$$w'(L) = \frac{C_1 L^2}{2} + C_2 L = 0 \Rightarrow C_2 = -\frac{C_1 L}{2} = \frac{PL}{2EI}$$ Randvillkor (4)

③ Bestäm $w(L)$

$$w(x) = -\frac{Px^3}{6EI} + \frac{PLx^2}{4EI}$$

$$\boxed{w(L) = \frac{PL^3}{12EI}}$$